Preprints.org (www.preprints.org) | NOT PEER-REVIEWED | Posted: 6 October 2025 doi:10.20944/preprints202509.2421.v2

3.2. Normalized Spurion Direction and Canonical Rank-One Tensor

It is convenient to record the projected spurion scalar and the normalized direction of $\partial\epsilon$:

$$\Sigma_\epsilon \;\equiv\; \Pi_{\rm PT}\big[(\partial\epsilon)^2\big], \qquad \sigma_\epsilon \equiv \mathrm{sgn}(\Sigma_\epsilon), \qquad \hat{n}_\mu \equiv \frac{\partial_\mu\epsilon}{\sqrt{|\Sigma_\epsilon|}} \quad (\Sigma_\epsilon \neq 0). \tag{14}$$

From $\hat{n}$ we define a *dimensionless*, traceless rank-one tensor

$$\mathcal{T}^\mu{}_\nu \;\equiv\; \sigma_\epsilon\Big(\hat{n}^\mu\hat{n}_\nu - \tfrac{1}{4}\,\delta^\mu{}_\nu\Big), \qquad \mathrm{Tr}\mathcal{T} = 0, \quad \mathrm{Tr}(\mathcal{T}^2) = \tfrac{3}{4}. \tag{15}$$

The (dimension-one) torsion trace scale along $\hat{n}$ will be denoted

$$\tau \;\equiv\; 3\eta\,\sqrt{|\Sigma_\epsilon|}. \tag{16}$$

When $\Sigma_\epsilon \to 0$ we work on $\Sigma_\epsilon \neq 0$ patches and extend by continuity via a smooth regulator $\Sigma_\epsilon \to (\Sigma_\epsilon^2 + \varepsilon^2)^{1/2}$, $\varepsilon \to 0^+$. We emphasize the notational distinction: $\mathcal{T}_\mu$ is the projectively invariant *trace vector* (13), while $\mathcal{T}^\mu{}_\nu$ in (15) is a traceless rank-one *matrix* built from $\hat{n}$.

3.3. What the Projector Allows (Pre- vs. Post-Lock)

By Theorem 2 and the parity assignments of Section 2, the following quadratic monomials with at most one derivative per building block are PT-even and thus survive the scalar projector:

$$T^A{}_{BC}T_A{}^{BC}, \qquad (\partial\epsilon)^2, \qquad T_\mu\,\partial^\mu\epsilon.$$

The mixed bilinear $T_\mu\partial^\mu\epsilon$ is therefore *independent pre-lock*.[1] Once the Palatini–PT uniqueness map (C1; Section 4) or, equivalently, the trace lock is enforced,

$$T_\mu = 3\eta\,\partial_\mu\epsilon \quad\Longrightarrow\quad T_\mu\partial^\mu\epsilon \;=\; 3\eta\,\Sigma_\epsilon, \tag{17}$$

and the mixed bilinear collapses to the spurion scalar.

For later reference we summarize the fate of the monomials:

Monomial	PT	Projected fate (pre-lock $\to$ post-lock)
$T^A{}_{BC}T_A{}^{BC}$	even	survives $\to$ survives
$(\partial\epsilon)^2$	even	survives $\to$ survives
$T_\mu\,\partial^\mu\epsilon$	even	survives, independent $\to 3\eta\,\Sigma_\epsilon$
$\nabla\cdot(\cdots)$	—	improvement $\to$ improvement

3.4. Two-stage closure at one-derivative order

Allowing at most one derivative per building block and working to quadratic order, the projected, real, PT-even scalar sector closes in two stages:

$$\begin{aligned} &\text{Pre-lock basis: } I_1 \equiv \Pi_{\rm PT}\big[T^A{}_{BC}T_A{}^{BC}\big], \quad I_2 \equiv \Sigma_\epsilon = \Pi_{\rm PT}\big[(\partial\epsilon)^2\big], \\ &\qquad\qquad I_3 \equiv \Pi_{\rm PT}\big[T_\mu\partial^\mu\epsilon\big]; \\ &\text{Post-lock (C1/trace-lock): } I_3 \xrightarrow{T_\mu = 3\eta\,\partial_\mu\epsilon} 3\eta\,\Sigma_\epsilon, \text{ so the basis reduces to } \{I_1, \Sigma_\epsilon\}. \end{aligned} \tag{18}$$

The self-adjointness of $\Pi_{\rm PT}$ on real scalars (9) and A4 (boundary posture) justify the integration-by-parts steps implicit in (18).

[1] Before any relation between T_μ and $\partial_\mu\epsilon$ is imposed, no integration-by-parts identity reduces $T\cdot\partial\epsilon$ to a combination of T^2 and $(\partial\epsilon)^2$ up to improvements.

© 2025 by the author(s). Distributed under a Creative Commons CC BY license.